

NewsBot 2.0 — Advanced NLP News Intelligence System

Reflection Journal

Class Final Project

Nichelle Graf

ITAI 2373 – Natural Language Processing | Spring 2026

Professor Anna Devarakonda

May 2, 2026

Introduction

This project challenged me to design and implement an end-to-end Natural Language Processing (NLP) system that transforms raw text into structured insights. While I had prior exposure to individual NLP concepts, this was my first experience integrating multiple components into a cohesive, working pipeline.

The project required both technical execution and system-level thinking, which pushed me to approach problem-solving more strategically.

Individual Contributions

I independently developed the NewsBot 2.0 system, including:

- Designing the modular system architecture
- Implementing the NLP pipeline (preprocessing, classification, sentiment, topics, entities, summarization)
- Building the evaluation framework and interpreting performance metrics
- Creating the GitHub repository and documentation (README, technical documentation)
- Developing structured outputs that combine multiple NLP insights

I also focused on making the system understandable and well-documented, not just functional.

Challenges Encountered

One of the main challenges was managing the complexity of integrating multiple NLP components into a single workflow. Each part of the system worked independently, but ensuring they functioned together required careful organization and troubleshooting.

Another challenge was working with development tools such as Jupyter Notebook, GitHub, and environment setup. Issues with formatting, execution flow, and repository structure required time and persistence to resolve.

Additionally, some NLP concepts—particularly topic modeling and evaluation—required deeper understanding to interpret results meaningfully rather than just generating output.

Problem-Solving Approach

To overcome these challenges, I broke the project into smaller, manageable parts and focused on one component at a time. I tested each section independently before integrating it into the full pipeline.

I also used iterative refinement—running the system, identifying issues, and improving the structure step by step. This helped me build confidence in both the code and the overall system design.

When I encountered technical issues, I worked through them methodically, using available resources and troubleshooting strategies rather than skipping over problems.

Lessons Learned

This project reinforced several key lessons:

- Building a working system is different from understanding isolated concepts
- Organization and structure are critical in larger projects
- Clear documentation is just as important as functional code
- Iterative development leads to better results than trying to perfect everything at once

I also gained a better appreciation for how NLP techniques work together in real-world applications.

Skills Developed

Through this project, I strengthened my skills in:

- Natural Language Processing concepts and applications
 - Data preprocessing and feature extraction (TF-IDF)
 - Machine learning implementation (Logistic Regression)
 - System design and modular architecture
 - Debugging and troubleshooting
 - Technical documentation and communication
-

Future Improvements

If I were to continue developing this project, I would focus on:

- Integrating more advanced models such as transformer-based approaches (BERT, GPT)
 - Improving topic modeling performance and interpretability
 - Adding real-time data integration using APIs
 - Enhancing the conversational interface for more dynamic interaction
 - Deploying the system as a web-based application
-

Personal Reflection

This project pushed me outside of my comfort zone and required persistence, especially when dealing with technical challenges and unfamiliar tools. There were moments where the process felt overwhelming, but working through those challenges helped me build both technical skills and confidence.

Completing this project gave me a better understanding of what it takes to build a real-world system, not just complete assignments. It also showed me that I am capable of working through complex problems step by step and producing a meaningful result.